

The impact of service principal (service robot vs. human staff) on service quality: The mediating role of service principal attribute

Shengliang Zhang^a, Zhenran Hu^a, Xiaodong Li^{b,*}, Ai Ren^c

^a School of Management, University of Science and Technology of China, Hefei, China

^b School of Economics and Management, Anhui Polytechnic University, Wuhu, China

^c School of Business, State University of New York at New Paltz, New Paltz, USA

ARTICLE INFO

Keywords:

Service principal
Attribute
Service robot
Service quality

ABSTRACT

In the context of the wide application of service robots, customers will question whether service robots or human staff provide higher service quality. Grounded in means-ends chain theory, this study explores the difference in SERVQUAL and in the driving mechanisms of different principals (i.e., service robots and human staff). The proposed model is examined through two scenario simulation experiments (restaurant and hotel contexts). The results show that reliability, responsiveness and empathy are influenced by different service principals (Study 1) and that the service principal's attributes mediate the relationships between the service principal and the dimensions of service quality (i.e., reliability, responsiveness and empathy) (Study 2). This study is the first to explore the specific differences in SERVQUAL from the perspective of the service principal. The findings may aid managers in selecting and using service robots more reasonably and help different service principals cooperate well.

1. Introduction

With the rapid development of technology, specifically artificial intelligence (AI) and robot manufacturing technology (Chiang & Trimi, 2020), service robots are increasingly being manufactured for and adopted in service industries (Jiang & Wen, 2020; Tung & Law, 2017). These robots are gradually replacing frontline employees as service principals in many service contexts, such as restaurants (Lu et al., 2021; Zhang et al., 2022; Zhu & Chang, 2020), hotels (Lee et al., 2020; Rosete et al., 2020), airports and train stations (Fuentes-Moraleda et al., 2020), banks, and hospitals (Xiao & Kumar, 2021). Their increasing adoption helps ensure service delivery without intimate contact between people (Ivanov et al., 2020).

While introducing service robots to replace human staff is a popular trend, service robots do not successfully deliver consistent service quality. On the one hand, service robots are considered to enhance customers' service experiences (Qiu et al., 2020). On the other hand, consumers prefer human staff to service robots as the latter cannot ensure adequate service quality (Yu & Ngan, 2019). For instance, several restaurants have had to 'fire' their robot staff because of poor service quality (Molly, 2016). Therefore, in order to successfully introduce robots to replace human staff in service encounters, it is critical to

elucidate the difference in service quality between service robots and human staff.

Recent studies have begun to compare the service provided by human staff and service robots. For example, in order to examine variance among different service principals (human staff and service robots), Byrd et al. (2021) drew on the gap model of service quality as the theoretical framework to construct a service gap model. Choi, Choi, Oh, and Kim (2020) noted that service robots are good at performing mechanical and repetitive tasks but in terms of interaction quality, robots will obviously perform worse than human staff. Although previous studies have examined the difference in service quality between service robots and human staff, it is still unclear what specifically drives this difference. Attributes of the service principal, which would influence service quality, have been mentioned in previous studies (Blut et al., 2015; Kano et al., 1984; Xiao & Kumar, 2021; Xu et al., 2017). As service principal attributes are the key to enabling an enterprise to beat its competitors in the market (Tan & Pawitra, 2001), the improvement of key attributes can maximize the advantages of service robots and better guarantee the efficient allocation of the business's resources (Lin et al., 2010; Tontini et al., 2007). Some key attributes of human staff have been developed and improved gradually to increase service quality and innovation (de Brentani, 2001; Keeling et al., 2010). It would be equally

* Corresponding author.

E-mail address: lixiaodong@ahpu.edu.cn (X. Li).

<https://doi.org/10.1016/j.jhtm.2022.06.014>

Received 31 January 2022; Received in revised form 21 June 2022; Accepted 22 June 2022

Available online 29 June 2022

1447-6770/© 2022 The Authors. Published by Elsevier Ltd. on behalf of CAUTHE - COUNCIL FOR AUSTRALASIAN TOURISM AND HOSPITALITY EDUCATION. All rights reserved.

effective to identify the driving mechanism of service quality difference based on the attributes of the service principal.

This study aims to build a conceptual framework rooted in means-chain (MEC) theory. We carried out two empirical studies through scenario simulation experiments. Study 1 was an experiment in the restaurant context to compare service quality provided by a service robot and human staff. Based on the findings of Study 1, Study 2 developed an experiment in the hotel context to examine the mediation mechanism of the relationships between service principal and service quality.

This study makes three contributions. First, it pioneers the elucidation of the difference of service quality between a service robot and human staff from the perspective of the service principal. Second, it is the first to figure out the key attributes of the service principal and, through investigation of those attributes, it explains the mediation mechanism of the difference in service quality between a service robot and human staff. Third, our study helps managers decide how to arrange service robots more efficiently and adjust the relationship between a service robot and human staff.

2. Literature review

2.1. Service robots

In 2016, a service robot was defined by the International Federation of Robotics as “a robot that performs useful tasks for humans or equipment excluding industrial automation applications.” Since then, this term has been pervasively used to describe the robots adopted in service contexts, which have been the potential service principals to replace human staff in service delivery roles. At present, more and more service industries, especially the hospitality and tourism sectors, have widely introduced service robots to replace all or part of the services previously provided by humans (Tuomi et al., 2021). For example, the Dadawan restaurant in the Netherlands has arranged for service robots to take over common tasks provided by human staff, such as serving food and cleaning up tables after dinner (Scheepers & Streukens, 2022). Restaurants in several countries are already using robotic chefs to replace human chefs (Zhu & Chang, 2020). Berezina et al. (2019) has described three types of tasks performed by robots in a restaurant: (1) loading and unloading the dishwasher; (2) chopping and preparing dishes; and (3) carrying food and drink from the kitchen to the guest's table. Service robots can also stand at the front of the restaurant and greet customers (Choi, Choi, Oh, & Kim, 2020). In shopping malls, service robots with mobile and conversational abilities can provide route guidance service and build rapport with customers (Hou et al., 2021). Robotic hotels not only use service robots to complete basic service tasks such as greeting guests, cleaning rooms and check-in/check-out services (Lee et al., 2020; Rosete et al., 2020), but in the tourism context, they also arrange the robots to perform social-emotional tasks such as answering visitors' complex questions and providing customized services (Lin et al., 2020; Wirtz et al., 2018).

Previous studies have revealed that customers have many differing perceptions and evaluations of service robots. On the one hand, service robots with professional abilities and unique value are considered to improve service performance (Huang & Rust, 2021; Zalama et al., 2014). Service robots have the ability to implement their work efficiently and correctly over long periods of time, so customers rate them as having faster service speed and higher service quality (Ivanov et al., 2020; Vatan & Dogan, 2021). In addition, the automated and contactless service provided by robots can increase customers' perceived safety (Ab Rahman et al., 2022). The entertainment value of service robots is also a key factor affecting service outcomes (Morita et al., 2018) because the way robots serve is often interesting and novel, thus bringing customers a sense of enjoyment and pleasure (Lin et al., 2020).

On the other hand, often service outcomes are not achieved because customers think service robots have low technical functions and lack

effective social interaction (Ghazizadeh et al., 2012; Hengstler et al., 2016; Wu et al., 2014). Since human-robot interaction has important implications for the customer experience (Choi, Choi, Oh, & Kim, 2020; Fuentes-Moraleda et al., 2020), service robots limited by their technology cannot have efficient human-robot interactions, which may lead customers to feel unsatisfied and isolated (Xiao & Kumar, 2021). Lee and Kim (2022) pointed out that currently, customers are not familiar with service robots, and thus they may think that the efficiency and convenience of robot service are relatively low.

Despite the fact that service robots have reshaped the essence of service (Belanche et al., 2020) and play a central role in the process of service delivery, whether the substitution of service robots for human staff as service principals improves performance (especially service quality) is still unclear, and there are very few empirical studies related to this question. We thus aim to examine the impact of service principal (service robot vs. human staff) on service quality.

2.2. Service quality

Previous research has revealed that service quality reflects the performance of the entire service system from the customer perspective (Lai, 2004; Oghuma et al., 2016). Among different frameworks for understanding the dimensions of service quality, SERVQUAL, proposed by Parasuraman et al. (1988), is widely accepted and pervasively adopted in many contexts. For example, Shah et al. (2020) used SERVQUAL to measure airline service quality based on data collected from international airline passengers. To evaluate the service quality of CQA (community question-answering) websites, Ming et al. (2021) constructed a CQA website service quality model based on the SERVQUAL model. Singh and Prasher (2017) applied the SERVQUAL model to a healthcare service quality evaluation system and then ranked the hospitals from the patients' perspective. SERVQUAL has been used not only for human staff, but also for other service principals associated with the new technology, such as web ports and information systems (Jiang et al., 2002; Li et al., 2002; van Iwaarden et al., 2003). The SERVQUAL scale has good universality and is regarded as a general and useful instrument to measure service quality in various service contexts (e.g., hospitality, tourism and restaurants) (Chand, 2010; Ding et al., 2020; Tzeng & Chang, 2011). Therefore, SERVQUAL is also suitable for measuring robot service quality.

Limited studies have examined service quality in the context of robot services. For example, through an empirical study, Chiang and Trimi (2020) ranked the importance of SERVQUAL's five dimensions in the order of assurance, reliability, responsiveness, tangibility and empathy. Morita et al. (2018) added two new dimensions (i.e., interactivity and entertainment) to SERVQUAL to measure robot service quality, while Kim and Lee (2014) simplified service quality into three dimensions (i.e., tangible quality, motion quality and system quality), which affect satisfaction through perceived usefulness. Prentice and Nguyen (2021) developed a four-dimensional scale for measuring robotic service quality (i.e., automatization, personalization, precision, and efficiency). Chen et al. (2022) used a mixed-method approach to develop a seven-dimensional scale that measures AI chatbot service quality (i.e., semantic understanding, close human-AI collaboration, human-likeness, continuous improvement, personalization, culture adoption, and efficiency). In studies of service robots per se in hospitality, some studies have revealed that service quality is influenced by elements of a service robot such as appearance, morphology, voice, expression and actions (Lu et al., 2021; Murphy et al., 2017, 2019; Shin & Jeong, 2020; Yoganathan et al., 2021). Xiao and Kumar (2021) found that the degree of robotics adoption influences service quality in a way that corresponds to an inverted U-shaped graph, while the nature of the company, service characteristics and brand equity play moderating roles.

As for the comparison between a service robot and human staff in terms of service quality, Wirtz et al. (2018) was the first to make some arguments. They noticed that service robots may deliver higher service

quality than human staff in routine service encounters, while human staff may deliver higher service quality than service robots in professional service roles (e.g., a divorce lawyer or a surgeon) due to their out-of-the-box thinking and creativity (Wirtz et al., 2018). Similarly, some studies have indicated that service robots are superior to humans when performing routine tasks but provide lower service quality than human staff due to technical limitations (Caic et al., 2019; Xiao & Kumar, 2021). Four experimental studies implemented by Kim et al. (2021) demonstrated that customers have a more positive attitude toward robot-staffed (vs. human-staffed) hotels in the context of the COVID-19 pandemic. Chan and Tung (2019) compared the service delivery of human staff and a service robot in a hotel and found that the service robot (vs. human staff) can provide a richer experience and emotional attachment for customers. Through a scenario simulation experiment, Choi, Choi, Oh, and Kim (2020) made a comparison of service quality among service robots, human staff and a combination of human staff and service robots; they found that human staff provided the highest interaction quality but that no significant difference in outcome quality was observed, and in terms of the effect on the physical service environment, human staff was perceived as slightly higher quality than a service robot.

Though these previous studies have enhanced our understanding of robot service quality, there are at least two limitations. First, it is still unclear what drives the specific difference in service quality between a service robot and human staff. Although some conceptual arguments (e.g., Caic et al. (2019); Wirtz et al. (2018)) and some comparisons of customer perceptions (e.g., Kim et al. (2021); Chan and Tung (2019)) have been made, the empirical evidence necessary to make a comparison between a service robot and human staff is still limited. To the best of our knowledge, only Choi, Choi, Oh, and Kim (2020) has provided empirical evidence for this difference, but this existing study does not reveal a possible driving mechanism. Second, there is no research that examines service quality from the perspective of the service principal per se, especially when introducing a new principal (e.g., service robots) to replace human staff. As focusing on the service principal in service encounters is more effective in ensuring the success of service design, it is necessary to identify the driving mechanism of service quality difference based on the attributes of the service principal. Therefore, this study aims to identify the driving mechanism and explain the difference in service quality from the perspective of the service principal.

2.3. Service principal and associated attributes

2.3.1. Service principal

The service principal, the customer and the service interaction are considered the three key elements of service performance. Service principal refers to the performer on one side of the bilateral interaction in a frontline service scenario (Zeithaml et al., 2008). It is the touchpoint of the service delivery system and represents the means by which service companies make contact with customers (Ho et al., 2020).

Previous studies have examined human staff as the service principal in service encounters and enhanced our recognition of how the service principal influences service quality. For example, when customers evaluate service quality, they take the performance of human staff as an important basis (Lai, 2004). Kumar et al. (2019) stated that a service enterprise can enhance customer experience by improving the emotional and interactive ability of human staff. As service robots have undergone rapid development and deep deployment, they have become an important service principal in the service industry (Tzou & Su, 2009).

2.3.2. Attributes of the service principal

Attributes refer to all the characteristics or features demonstrated by a service principal and perceived by users (Elliott et al., 2012). The differences in service functions and performance among service principals are derived from variation in attributes (Tung & Law, 2017). Previous studies have revealed several typical attributes of human staff in

service encounters (see Table 1). For example, initiative is an intrinsic incentive attribute that promotes friendly communication between human staff and customers and that can help obtain customers' trust (Frese et al., 1997; Keeling et al., 2010). Recent research has also identified some attributes of service robots (see Table 1). For instance, anthropomorphism is a key attribute related to the appearance of a service robot that affects customers' emotional response to it (Bartneck et al., 2009; van Doorn et al., 2017). Although previous studies have mentioned the obvious and typical attributes of a service principal, they are scattered, and no study has summarized the key attributes of a service principal, especially those attributes associated with service quality.

According to the seller-customer communication model developed by Sheth (1976), a service principal can be represented by three dimensions of attributes, i.e., task-orientation, interaction-orientation and self-orientation. Task-orientation relates to the work abilities and professional spirit of the service principal and is reflected through stability and absorption. Stability is considered one of the most important attributes in different contexts, as it determines service delivery and service failure attribution (Tuli et al., 2007; Tung & Law, 2017). Previous research reveals that stability is the basic requirement for service tasks accomplished by either a service robot or human staff (de Brentani, 2001; Tuli et al., 2007). Meanwhile, absorption is the manifestation of individuals' work attitude and can reflect what attitude a service principal adopts in dealing with tasks and customers (Schaufeli et al., 2002). Ivanov et al. (2018) has indicated that robots demonstrate absorption during the service process, which inspires customers' trust. As for human staff, absorption during the service process will motivate them to take the initiative to learn and improve their work ability (Sonntag & Kruegel, 2006). Thus, we choose stability and absorption to represent the dimension of task-orientation.

Interaction-orientation refers to how a service principal interacts with customers and includes adaptability and friendliness (Keeling et al., 2010; Köhler et al., 2011). Previous studies have found that adaptability, which refers to adjustment to the provisional needs of customers and unpredictable conditions, reflects a service principal's ability to assess and solve emergent problems (Fugate et al., 2004). Adaptability has

Table 1
Attributes of service principals.

Service principal	Key attribute	Source
Human staff	Stability	Tuli et al. (2007)
	Adaptability, interaction	Teng and Liu (2014)
	Absorption	Sonntag and Kruegel (2006)
	Creativity, initiative	de Brentani (2001)
Service robot	Friendliness, interaction, initiative	Keeling et al. (2010)
	Stability	de Brentani (2001)
		Tung and Law (2017)
	Adaptability	Kabadayi et al. (2019)
		Tung and Law (2017)
	Absorption	Chi et al. (2021)
	Creativity	Verhagen et al. (2014)
	Friendliness	van Doorn et al. (2017)
	Anthropomorphism, continuity, likeability, animacy, perceived intelligence, perceived safety	Bartneck et al. (2009)
	Intelligence, predictability, adaptability	Berezina et al. (2019)

Table 2
Profile of the participants in Study 2 (N=222).

Variable		Frequency	Percent (%)
Gender	Male	96	43.2
	Female	126	56.8
Age	20 and under	5	2.3
	21–25	129	58.1
	26–30	39	17.6
	31–40	38	17.1
	Above 40	11	5.0
Education	High school/Below high school	5	2.3
	Junior college	15	6.8
	Undergraduate	106	47.7
	Graduate and above	96	43.2
Monthly expenses	Less or equal to 1000 RMB	14	6.3
	1001–3000 RMB	139	68.6
	3001–6000 RMB	33	16.9
	More than 6000 RMB	36	8.2

been identified as one of the core attributes for both human staff and service robots (Burns et al., 2018; Kabadayi et al., 2019). Meanwhile, friendliness creates a sense of mutual understanding between a service principal and customers (Price et al., 1995). van Doorn et al. (2017) argued that friendliness is an important attribute that reflects social existence and increases customers' interest and warmth. For human staff, friendliness is vital in establishing interpersonal relationships with customers (Keeling et al., 2010). Thus, we choose adaptability and friendliness to represent the dimension of interaction-orientation.

Self-orientation is the tendency of individuals to strive to maintain and improve their abilities. Creativity shows that a service principal is undertaking self-improvement to improve individual innovative ability (Zhou & George, 2001). Previous studies have discovered that the level of creativity that a service robot exhibits directly affects its acceptance by customers (Pozharliev et al., 2021; Reese, 2018). For human staff, creativity is an important attribute that requires an innovative spirit and actions (de Brentani, 2001). Thus, we choose creativity to represent the dimension of self-orientation.

Meanwhile, the five proposed attributes (i.e., stability, absorption, adaptability, friendliness, creativity) have been fragmentarily examined in previous studies (see Table 1). They are mentioned as the key mutual attributes of different service principals (human staff and service robots). Thus, it is reasonable to choose these five attributes to describe a service principal.

2.4. Proposed model

We adopt MEC theory to elucidate the effects of a service principal (a service robot or human staff) on service quality through attributes. MEC theory is a framework proposed by Gutman (1982) that is used to explore the key reasons why customers choose a service. It reflects how consumers achieve their desired consumption goals from a service provider.

According to MEC theory, means refers to an approach to achieving a goal (i.e., ends), and this approach is conducted through attributes of an object. As consumers regard each product as a collection of attributes to fulfil their needs (Kotler, 1994), they similarly consider a service principal as a set of attributes to interact with them and deliver their requested services (Stanton et al., 1991). Considering that service quality is generally regarded as the consequence and value (i.e., ends) conveyed by a service principal (through their delivery process) (Blut et al., 2015), we thus propose the driving mechanism to explain the difference in service quality from different service principals (i.e., service robot vs. human staff) (see Fig. 1).

Specifically, we divide our problem into three parts (i.e., service principal, attributes, and service quality) with the assistance of MEC theory. First, we distinguish service principals as a service robot and human staff. Then we adopt three dimensions of attributes as the means and SERVQUAL as the ends. We examine the relationships between service principal and service quality. In addition, consumer psychology shows that consumers are able to evaluate their experiences and feelings based on the attributes of the service principal and then form their subjective feelings about service quality (Gardial et al., 1994). Some studies have shown that the attributes of a service principal have a positive impact on user experience (Qiu et al., 2020). Both categories of prior studies indicate that our proposed model is reasonable.

3. Hypotheses

3.1. Service quality

3.1.1. Reliability

Because robots are controlled by a specific program, their behavior is highly stable, and there is no possibility that they will exhibit fatigue, sadness, irritability, or any other negative emotion. Unless a robot is malfunctioning, the service it performs will be completed exactly according to established procedures, resulting in a high level of service

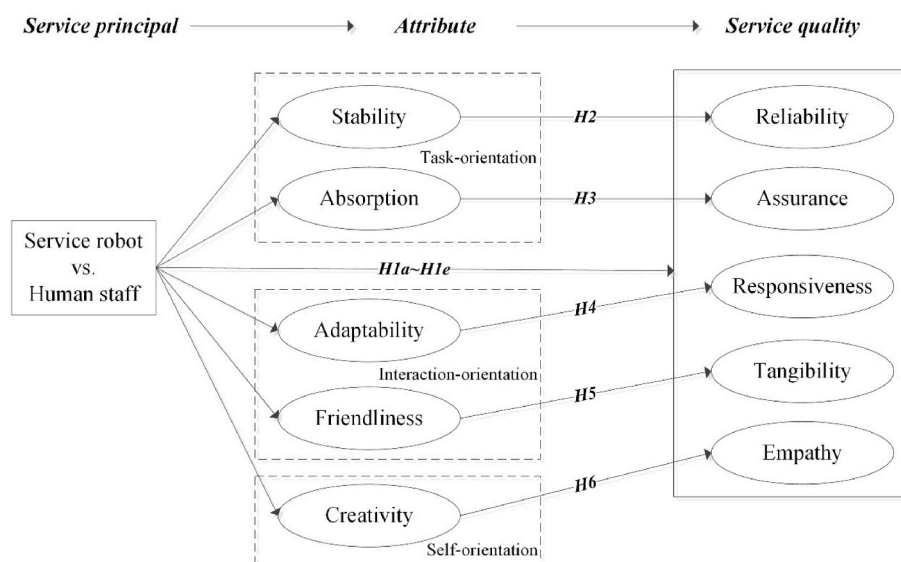


Fig. 1. Proposed model.

reliability (Reese, 2018). As human staff have physical limitations and can experience and exhibit a variety of emotions, inaction and burnout can occur. Humans tend to be unable to work consistently for a long period of time, thus diminishing their service reliability. On this basis, we propose Hypothesis 1a:

H1a. The service principal has an impact on reliability, i.e., the reliability of a service robot is higher than that of human staff.

3.1.2. Assurance

Assurance mainly measures the degree to which the service principal's ability inspires the customer's trust. Robots purchased by service enterprises naturally have basic service capabilities and can be directly put into service scenarios. The professional capabilities of robots have become diverse and mature with the continuous upgrading of artificial intelligence technology. However, human staff need constant training to truly master various service skills. For example, a service robot can deliver several customers' cutlery at a time, a workload equivalent to that of several human employees. When a customer asks for information about a dish while ordering, the robot can give a quick answer, whereas the human employee may be unable to answer or may give the wrong information. From the perspective of professional capabilities, then, service robots are better than human employees. In general, customers consider service robots as more efficient and trustworthy than human agents (Ivanov et al., 2018), leading some researchers to conclude that if full-service restaurants implemented service robots, the overall service flow would be more convenient and hassle-free (Lin & Mattila, 2021). Another relevant factor is that customers tend to be more tolerant of service robots than of humans, trusting them even when they make minor service mistakes (Reese, 2018). Hypothesis 1b is thus formulated as:

H1b. The service principal has an impact on assurance, i.e., the assurance of a service robot is higher than that of human staff.

3.1.3. Responsiveness

Service robots generally work according to pre-set programs. Although several robots can rapidly imitate and learn tasks, this must be based on complex data analysis (Wuenderlich, 2009). When facing emergencies or vague customer demands, robot comprehension is relatively poor, which can affect response speed. In contrast, humans are living organisms, generally receive professional training, and can demonstrate adequate understanding and adaptability in complex situations. Under normal circumstances, human staff can respond to customer requests and requirements to promptly and quickly provide appropriate services. Thus, Hypothesis 1c is stated as:

H1c. The service principal has an impact on responsiveness, i.e., the responsiveness of human staff is higher than that of a service robot.

3.1.4. Tangibility

Although many service robots have a human-like appearance, their appearance, expressions and movements are often still criticized as unnatural and creepy. For instance, when a robot moves its mouth, its facial expression does not change much, which makes customers feel uneasy (Guizzo, 2010), and for customers accustomed to human waiters, it is strange to see a robot moving around a restaurant. Others say the appearance, layout and decoration of many restaurants clashes with the mechanical appearance of the robots: the traditional restaurant has a warm and comfortable atmosphere, while robots are cold, inhuman machines that represent high-end technology. When the service scenario conflicts with the service subject, it is natural for the customer to perceive lower service quality. Thus, we propose:

H1d. The service principal has an impact on tangibility, i.e., the tangibility of human staff is higher than that of a service robot.

3.1.5. Empathy

Empathy is a social concept that encompasses social perspective-taking and understanding others' motivations, and it plays a critical role in shaping customer responses to a service principal (Wieseke et al., 2012). Past research shows that service robots (compared to human staff) are perceived as less socially expressive and less capable of understanding customers' feelings (Paiva et al., 2017). Many service companies originally intended to introduce service robots with the goal of replacing human employees in accomplishing boring and repetitive tasks. Robot manufacturers have also standardized the production of robots and strive to provide customers with homogeneous services. This creates a problem: the efficiency of social interaction between service robots and customers is relatively low (Wirtz et al., 2018). As a service provider, a service robot is seriously lacking in personalization, initiative and creativity, so the human-robot interaction (HRI) in restaurants does not have a real social nature. Although some service robots are highly intelligent and can have simple conversations with customers, this is far from normal social interaction (Pozharliev et al., 2021). Hypothesis 1e is thus stated as:

H1e. The service principal has an impact on empathy, i.e., the empathy of human staff is higher than that of a service robot.

3.2. Attributes of service principal

3.2.1. Stability

Heider and Fritz (1958) classified the stability of service personnel into three dimensions: the consistency between individual employees and an organization, the consistency of customers' perception of employees, and the consistency of individual employees over the entire service process. These dimensions have an obvious impact on the work efficiency of employees. Service robots follow specific algorithms and procedures; their goals are highly consistent with those of their employers, and they can provide very stable customer service over a long period of time. No matter how many times a robot delivers cutlery to the same customer, its performance will be consistent and high-quality, and the customer's perception will not change over time. However, maintaining such a high level of consistency is extremely difficult for human personnel. There are many factors that can affect the stability of human staff. Conflicts with the goals of the company or colleagues, negative emotions, and fatigue may make a human employee exhibit different service states in the same service link. A high level of stability indicates that as the possibility of change and uncertainty is reduced, the probability and degree of service failure is also reduced (Nowak et al., 2005). Tuli et al. (2007) found that the stability of service principals is positively correlated with their demonstrated work efficiency. Therefore, compared to human staff, service robots with a higher level of stability can better meet service standards, increasingly fulfil enterprises' service promises, and remain above service standards for a long period. This thinking brings us to Hypothesis 2:

H2. Stability mediates the relationship between service principal and reliability.

3.2.2. Absorption

Absorption is defined as the degree to which one concentrates fully and engrosses oneself deeply in one's work. It makes employees feel like time passes quickly, so that they do not want to be detached from their work, and it is a direct reflection of how engaged they are in their work (Schaufeli & Bakker, 2001). Absorption is an important antecedent of employee performance output. It can greatly improve efficiency and service quality in service encounters (Dane & Brummel, 2013). For a service robot, strict preset programs ensure that it can be isolated from external environmental or work distractions and immersed in the working atmosphere. A service robot can perform specific service tasks and maintain high absorption continuously (Reese, 2018). This makes customers feel that the service robot is professional, confident and

decisive. Thus, customers believe that the human-machine interaction will have positive service results (Chi et al., 2021). As for human staff, they cannot maintain high absorption for long periods of time. They may suffer from physical discomfort, face conflicts between professional roles and personal characteristics, or even develop prejudices against certain customers that could hinder them in providing consistent customer service. These factors can all affect service outcomes (Richard & Mansfield, 1996). Some studies revealed that service robots with higher absorption are better situated and have more opportunities to gain customers' positive evaluation of their work abilities (Reese, 2018). Based on this evidence, we propose Hypothesis 3:

H3. Absorption mediates the relationship between service principal and assurance.

3.2.3. Adaptability

Some surveys have queried people about the tasks that service robots should perform (Carpenter et al., 2009), finding that service robots can perform a wide variety of tasks. Particularly in the catering service industry, customers' personalized requirements are quite wide-ranging. A service principal may need to address questions and complaints, give drinks, plates, and tablecloths to customers, adjust the temperature of the air conditioner, and listen to customers' suggestions. Tackling these numerous tasks requires service robots to have the flexibility to adapt to changing and sometimes novel task constraints (Jones & Schmidlin, 2011). Adaptability helps convey a human-oriented experience and gains customers' favor in a short time. The adaptability level of a robot relies very highly on programming and data. Currently, robots in catering services are often not sufficiently adaptable, are unable to make quick judgments in the face of customer requirements, and perform poorly in terms of responsiveness (Tung & Law, 2017). In contrast, human employees are highly adaptable, which helps them gain experience and respond quickly to external changes and customer demands. Compared with service robots, human staff adapt better based on changing customer and situational inputs (Kabadiyi et al., 2019). Therefore, we propose the following:

H4. Adaptability mediates the relationship between service principal and responsiveness.

3.2.4. Friendliness

Customers may also more readily feel affinity and attachment to human staff than to human-like robots due to the perceived similarity. This lower affinity and lower sense of friendliness may make customers think that service robots appear more like machines than humans (Mori, 2012). A low sense of friendliness can also make customers perceive the appearance of a robot as threatening or frightening, which often leads to a bad first impression (Ackerman, 2016). Service robots display different degrees of anthropomorphism, and when the degree of anthropomorphism is too low and a robot looks completely unlike a human being, this gives people a feeling that it is something dead, and customers will see it as just a machine that can make sound. Yet when the degree of anthropomorphism is too high and a robot's appearance and voice are quite similar to a human being's, people will feel fear and anxiety (Xiao & Kumar, 2021) and may even have an 'uncanny valley' experience (Mori, 2012). In short, due to the lack of affinity in the appearance of service robots, people have a poor perception of their tangibility. Human staff, especially those in direct contact with customers, generally have a dignified appearance, decent dress, a cordial expression, and an amiable attitude, and they show personalized care for customers through smiles and greetings. Therefore, customers feel the warmth and friendliness of human staff and are more willing to approach them. Accordingly, we propose:

H5. Friendliness mediates the relationship between service principal and tangibility.

3.2.5. Creativity

Creativity requires the service principal to produce innovative ideas and put them into practice (Reese, 2018). Since most service robots in restaurants are less creative, they can only complete the most basic service tasks. Meanwhile, customers not only want professional service, but they also want to enjoy the humanistic care of service providers. Therefore, the creativity of the service principal influences customers' perceived service quality. Human staff who are more creative have an advantage in the empathic dimension. For example, when a waiter in a restaurant finds that a customer is unhappy after delivering their food, they may give the customer snacks and drinks and ask if they are angry about the food or the service. However, in the same situation, a service robot would directly ignore the customer's emotional changes (Hu et al., 2021). In addition, due to technical limitations, the intelligence level of robots currently used in the service industry, especially the catering industry, is generally not high enough. Without emotion and empathy, these robots can only provide services for customers mechanically in accordance with pre-set procedures. Given this lack of creativity, service robots fail to identify and meet customers' personalized requirements (Reese, 2018). Human staff, in contrast, are living organisms with thoughts and empathy. At the same time, most service enterprises encourage human staff to creatively provide personalized services for customers, so human staff can understand, identify and strive to meet customers' personalized needs. Xiao and Kumar (2021) found that in restaurant service, human employees are more creative than service robots and have better insight into customers' needs, thus performing higher quality service. As such, we put forth the following hypothesis:

H6. Creativity mediates the relationship between service principal and empathy.

4. Study 1

4.1. Procedure and sample

Study 1 accomplished its objective by means of scenario simulation (simulating a real scene, guiding the subjects into the scene and giving them a realistic experience) and a questionnaire survey. This study is set in the food service context for three reasons. First, the food service industry is a typical service industry. Second, the use of service robots is relatively common in the industry. Third, food service is a very familiar scene, so it would be easy for participants to have a realistic experience. A pre-test of 39 respondents was conducted prior to actual data collection to ensure the flow and clarity of the questionnaire.

Data were collected during August–September 2020. We recruited respondents via social media platforms. After obtaining their permission, we distributed the questionnaire (including scenario simulation) through wjx.com and kept in touch with them through Tencent Meeting until they finished the experiment. Only customers who met two criteria were recruited: participants must be over 18 and must have visited a restaurant with robot service in the past six months. In order to promote acceptance of the survey and high-quality responses, all participants received a cash reward of ¥10. We obtained 305 valid (response rate of 82.4%) responses. Among the respondents, 165 are female (54.1%) and 140 are male (45.9%). Their ages range from 19 to 53, with an average of 29.6.

4.2. Study design

Scenarios were designed for the experiment. Considering that in the marketing literature, it is common to examine customers' perception upon exposure to scenarios as experimental stimuli (Choi, Choi, & Mattila, 2020), we designed two scenarios, one of robot service and one of human service. Participants would randomly select one of the two scenarios (simulated in the Appendix) to have an immersive experience. Each scenario was composed of three service encounters: reception,

ordering and serving. Pictures and text were used to describe the scene in detail, as the two were considered as the effective materials to characterize a specific situation (Choi, Choi, Oh, & Kim, 2020). To further ensure that participants understood the situation and were able to report their true feelings and to address any possible confusion, research assistants remained online while participants went through the scenario.

Based on their true feelings arising from the scenario, participants were then asked to complete the questionnaire, which included the five dimensions of service quality and respondents' information (i.e., gender, age, educational background and average monthly expenses).

4.3. Measures

The items used to measure five dimensions of customers' service quality were adapted from the scale developed by Parasuraman et al. (1988). All the constructs were measured using a Likert scale ranging from 1 (strongly disagree) to 7 (strongly agree).

To ensure the validity of the measures, seven people, including two associate professors, two doctoral students from the department of marketing, and three customers, were invited to modify item wording to fit the catering service scenarios. The back-translation technique was also adopted to translate all items between English and Chinese. A pre-test with 39 responses eventually formed the measures comprising 17 items (four items for reliability, four items for responsiveness, and three each for the other three variables) and showed the questionnaire was logically consistent and easy to understand.

We controlled for the effects using an unmeasured latent methods factor (ULMC) technique to examine potential common method bias (Richardson et al., 2009). Using AMOS22.0, we constructed a confirmatory factor analysis (CFA) model (M1) and then constructed a model M2 containing a common method factor. The comparison results of the main fitting indexes between M1 and M2 were $\Delta CFI = 0.005$, $\Delta TLI = 0.007$, $\Delta SRMR = 0.021$, and $\Delta RMSEA = 0.016$. As the difference of each fitting index was less than 0.03 (smaller than 0.05 cut-off), it indicated there was no obvious common method bias in this study.

According to M1, results showed that the data and the measurement model fit well with $\chi^2/df = 1.109$, RMSEA = 0.017, CFI = 0.995, IFI = 0.955 and TLI = 0.993. Cronbach's α values of reliability, assurance, responsiveness, tangibility and empathy were 0.897, 0.847, 0.823, 0.660 and 0.891, which indicated that the reliability of the scale was acceptable.

4.4. Results

To compare service quality of human staff and a service robot, an independent sample *t*-test was conducted (Fig. 2). As expected, the

reliability of a service robot was significantly higher than that of human staff ($M_{SR} = 5.403$, $M_{HS} = 3.818$, $t(303) = -9.884$, $p < 0.001$). The responsiveness ($M_{SR} = 3.969$, $M_{HS} = 5.278$, $t(303) = 8.927$, $p < 0.001$) and empathy ($M_{SR} = 3.364$, $M_{HS} = 5.201$, $t(303) = 11.346$, $p < 0.001$) of human staff were significantly higher than that of a service robot. Therefore, H1a, H1c, and H1e were supported. However, for assurance ($M_{SR} = 4.991$, $M_{HS} = 5.139$, $t(303) = 0.980$, $p > 0.05$) and tangibility ($M_{SR} = 5.071$, $M_{HS} = 5.199$, $t(303) = 1.122$, $p > 0.05$), no significant differences were observed. These findings disproved H1b and H1d.

5. Study 2

Study 2 was designed to examine the mediation mechanism of the relationships between service principal and service quality.

5.1. Procedure and sample

In Study 2, as hotels are popular venues for the application of service robots (Chiang & Trimi, 2020; Lin & Mattila, 2021), we replaced the restaurant scenario in Study 1 with the hotel scenario, the materials of which were adapted from the study by Choi, Choi, Oh, and Kim (2020) (see Appendix). The experiment procedure and the recruitment criteria of Study 2 were same as in Study 1. The experiment lasted for 2 months at the end of 2020. A total of 356 participants were recruited, and 222 valid responses were finally collected (see Table 2).

5.2. Measures

All the constructs were measured using a Likert scale ranging from 1 (strongly disagree) to 7 (strongly agree). Service quality was measured the same way as in Study 1. As for the five attributes of the service principal, items for stability were adapted from Hess et al. (2003), absorption was examined using the scale developed by Schaufeli et al. (2002), adaptability was evaluated based on the instrument used by Briscoe et al. (2006), items of friendliness were drawn from Tsai and Huang (2002), and creativity was estimated by the measurement created by Tierney et al. (1999). We adopted the same item polishing and pre-test procedures used in Study 1 to ensure the validity of the measures.

The UMLC technique was adopted to examine potential common method bias, as in Study 1. The comparison results of the main fitting indexes between M1 (CFA model) and M2 were $\Delta CFI = 0.021$, $\Delta TLI = 0.022$, $\Delta SRMR = 0.012$ and $\Delta RMSEA = 0.007$. The difference of each fitting index indicated that there was no obvious common method bias in this study.

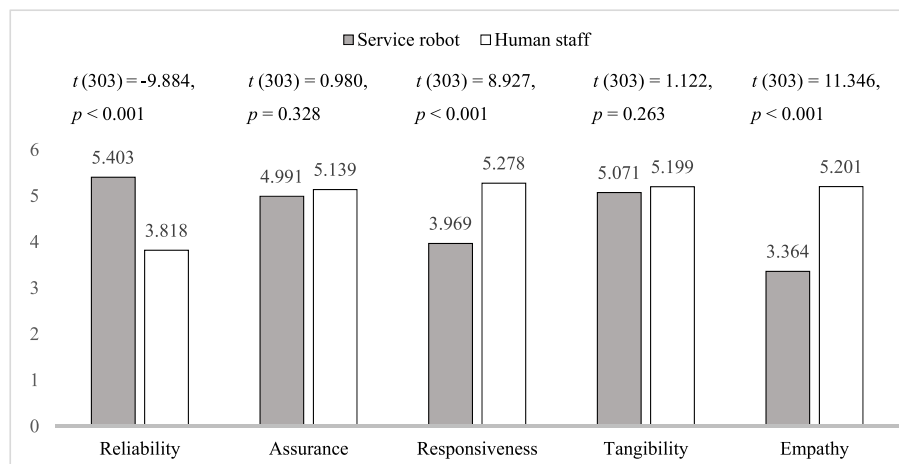


Fig. 2. Independent sample *t*-test analysis result.

5.3. Results

5.3.1. Measures assessment

We ran a CFA model with AMOS 22.0 and got the model fit indexes with $\chi^2/df = 1.934$, RMSEA = 0.065, CFI = 0.917, IFI = 0.918, TLI = 0.901. As Table 3 shows, the loadings for all items were above 0.5 and the values of composite reliability and Cronbach's α were all above 0.7; thus, the reliability of the ten constructs was good. Meanwhile, the AVE's square root of each construct exceeded the correlation value, which showed good discriminant validity (see Table 4).

5.3.2. Hypotheses test

Like in Study 1, we first compared the difference in service quality of the two service principals. The reliability of a service robot was significantly higher than that of human staff ($M_{SR} = 5.341$, $M_{HS} = 4.757$, $t(220) = -4.498$, $p < 0.001$). The responsiveness ($M_{SR} = 4.658$, $M_{HS} = 5.289$, $t(220) = -4.431$, $p < 0.001$) and empathy ($M_{SR} = 3.484$, $M_{HS} = 5.013$, $t(220) = 8.335$, $p < 0.01$) of human staff were significantly higher than that of a service robot. For assurance ($M_{SR} = 5.042$, $M_{HS} = 5.289$, $t(220) = 1.949$, $p > 0.05$) and tangibility ($M_{SR} = 4.691$, $M_{HS} = 4.994$, $t(220) = 1.929$, $p > 0.05$), no significant differences were observed. These results further confirmed the findings of study 1.

Next, following the mediation examination procedure from Baron and Kenny (1986) and the PROCESS method proposed by Hayes and Preacher (2014), we investigated the mediating effects of service principal attributes by running Process Macro in SPSS22.0.

As an example, the service principal had significant influences on stability ($a = 1.0580$, $p < 0.01$) and reliability ($c = 0.5842$, $p < 0.001$), while stability had a significant influence on reliability ($b = 0.5843$, $p < 0.001$). When we added service principal and stability at the same time as independent variables, the coefficient between service principal and reliability (dependent variable) became smaller but insignificant ($c^* = -0.0340$, $p > 0.05$), and the 95% confidence interval (Table 4) did not

include zero, indicating the existence of a mediation effect. Thus, H2 was supported.

Service principal was significantly related to adaptability ($a = -1.1242$, $p < 0.001$), and adaptability was significantly related to responsiveness ($b = 0.5066$, $p < 0.001$). Service principal had a significant influence on responsiveness within the direct effect model ($c = -0.6311$, $p < 0.001$). However, the relationship between service principal and responsiveness was not significant when including adaptability as a mediator ($c^* = -0.0616$, $p > 0.05$), and the 95% confidence interval (Table 6) did not include zero. Thus, this study suggested that adaptability is a mediator between service principal and responsiveness. This finding provided support for H4.

The service principal had significant influences on creativity ($a = -1.4542$, $p < 0.001$) and empathy ($c = -1.5293$, $p < 0.001$), while creativity had a significant influence on empathy ($b = 0.7399$, $p < 0.001$). When we added service principal and creativity to the model, the regression coefficient of empathy was reduced, but it still remained significant ($c^* = -0.4533$, $p < 0.01$) when creativity was included as a mediator. We concluded that creativity is a mediator between service principal and empathy. Therefore, H6 was supported.

In addition, we ran the mediation procedures and found that there were no significant regression coefficients between service principal and assurance ($p > 0.05$) or between service principal and tangibility ($p > 0.05$) (Table 5). The results confirmed the relationships revealed in the t -test and showed that there was no mediating effect between absorption and assurance or between friendliness and tangibility. Thus, H3 and H5 were not supported.

6. Discussion

6.1. Key findings

Based on means-ends chain theory, this study explores the influence of different service principals on service quality and further identifies the mechanism of influence. Through scenario simulation experiments and data analysis, we draw the two following conclusions.

First, the findings specifically describe the difference in service quality between service robot and human staff. The reliability of a service robot is higher than that of human staff, a conclusion that further verifies the view of Ivanov et al. (2020) that service robots are good at repetitive and more dangerous tasks in hotel settings. Customers rate the responsiveness and empathy of human staff as higher than that of a service robot, which is consistent with the results of Choi, Choi, Oh, and Kim (2020). This result supports previous arguments (Qiu et al., 2020; Wirtz et al., 2018) that service robots in restaurants and hotels have functional defects in their communication with customers and lack humanization, so they cannot completely replace human staff in some respects at present.

However, in the dimensions of assurance and tangibility, there is no significant difference in service quality between the two service principals. In terms of assurance, catering and hotel service work is relatively simple, and the three service links described in this study in particular are routine and repetitive tasks. Human staff have received professional training and their service error rate will be relatively low (Tung & Au, 2018), so the assurance of human service is not worse than that of robot service. As for tangibility, service robots are completely different from human beings in appearance and are thus unable to communicate with customers through facial expressions and body movements. It is also difficult to visually display sincere smiles and enthusiastic service attitudes like human employees do (Ariffin & Maghzi, 2012). As we learned from communicating with some of the respondents, a service robot's appearance can give customers a sense of novelty and stimulate their curiosity and interest. Human staff and service robots thus each have their own advantages in terms of external image, so there is no significant difference between them in terms of tangibility.

Second, these findings reveal the mediation effects of service

Table 3
Loading, construct reliability, and validity results.

Variables	Items	Loadings	AVE	Cronbach's α	C.R.
Reliability (RE)	RE1	0.662	0.478	0.789	0.785
	RE2	0.706			
	RE3	0.656			
	RE4	0.783			
Assurance (AS)	AS1	0.750	0.511	0.759	0.758
	AS2	0.670			
	AS3	0.722			
Responsiveness (RES)	RES1	0.830	0.518	0.743	0.761
	RES2	0.708			
	RES3	0.604			
Tangibility (TA)	TA1	0.827	0.604	0.812	0.820
	TA2	0.793			
	TA3	0.706			
Empathy (EM)	EM1	0.846	0.766	0.928	0.929
	EM2	0.883			
	EM3	0.906			
	EM4	0.864			
Stability (ST)	ST1	0.677	0.548	0.777	0.784
	ST2	0.800			
	ST3	0.739			
Absorption (AB)	AB1	0.747	0.648	0.847	0.846
	AB2	0.840			
	AB3	0.825			
Adaptability (AD)	AD1	0.737	0.629	0.831	0.835
	AD2	0.853			
	AD3	0.785			
Friendliness (FR)	FR1	0.750	0.687	0.861	0.868
	FR2	0.858			
	FR3	0.873			
Creativity (CR)	CR1	0.866	0.755	0.902	0.902
	CR2	0.881			
	CR3	0.859			

AVE: Average Variance Extracted; C.R.: Composite Reliability.

Table 4

Discriminant validity.

	1	2	3	4	5	6	7	8	9	10
Reliability	0.691									
Assurance	0.470**	0.715								
Responsiveness	0.163**	0.470**	0.720							
Tangibility	0.395**	0.402**	0.403**	0.777						
Empathy	0.055**	0.434**	0.558**	0.384**	0.875					
Stability	0.682**	0.326**	0.087	0.298**	−0.072	0.740				
Absorption	0.172**	0.464**	0.461**	0.290**	0.695**	0.035	0.805			
Adaptability	0.131**	0.468**	0.582**	0.470**	0.749**	0.013	0.715**	0.793		
Friendliness	0.269**	0.461**	0.496**	0.717**	0.559**	0.140**	0.513**	0.615**	0.829	
Creativity	0.023	0.306**	0.511**	0.356**	0.785**	−0.083	0.747**	0.764**	0.521**	0.869

Note: *p < 0.01. **p < 0.001.

Table 5

Mediating effect results (PROCESS MACRO) – regression coefficient.

Mediator	Independent variable	Dependent variable	Regression coefficient			
			a	b	c'	c
ST	SRvsHS	RE	1.0580*	0.5843**	−0.0340	0.5842**
AB	SRvsHS	AS	−1.1170**	0.3680**	0.2016	−0.2463
AD	SRvsHS	RES	−1.1242**	0.5066**	−0.0616	−0.6311**
FR	SRvsHS	TA	−0.7907**	0.5724**	0.2196	0.2169
CR	SRvsHS	EM	−1.4542**	0.7399**	−0.4533*	−1.5293**

Note: *p < 0.01. **p < 0.001. “SR” refers to service robot. “HS” refers to human staff.

Table 6

Mediating effect results (PROCESS MACRO) – 95% confidence interval.

Stage	Indirect effects	Indirect effects 95% confidence interval		Direct effects	Direct effects 95% confidence interval	
		lower limit	Upper limit		lower limit	Upper limit
SP→ST→RE	0.6180**	0.4191	0.8334	−0.0340	−0.2532	0.1852
SP→AB→AS	−0.4110**	−0.6092	−0.3012	0.2016	−0.0483	0.4516
SP→AD→RES	0.5695**	−0.7727	−0.3920	−0.0616	−0.3287	0.2056
SP→FR→TA	−0.4526**	−0.7573	−0.2959	−0.2169	−0.0096	0.4434
SP→CR→EM	−1.0760**	−1.3528	−0.8064	−0.4533*	−0.7407	−0.1659

Note: a.*p < 0.01. **p < 0.001.

b. Indirect and direct effects are calculated using bootstrap 5000 results. “SP” refers to service principal.

principal attributes on the relationship between service principal and service quality. Stability mediates the relationship between the service principal and reliability. This finding shows that in the service industry, especially in the service links and service tasks that require stability, it is completely feasible to let service robots replace human staff because it may improve service quality. Further, we find that adaptability mediates the relationship between the service principal and responsiveness while creativity plays a mediating role in the relationship between the service principal and empathy. These two conclusions verify the findings of Huang and Rust (2021) that service robots lack the thinking and analytical ability needed in the service process and have very low communication efficiency with customers.

However, absorption does not mediate the relationship between the service principal and assurance. The main reason may be that absorption of the service principal at present has a weak influence on delivering or accomplishing a service. Like some respondents indicated, consumers generally trust the working abilities of both human staff and robots and think the service principal can accomplish the assigned service tasks, although service robots are more attentive than human staff. Consumers particularly understand and accept the occasional distraction of human staff and think that their low absorption has no impact on assurance of a specific service delivery. Similarly, friendliness fails to mediate the relationship between the service principal and tangibility. Although the service principal has an impact on friendliness, the automated social presence that service robots possess will help to increase customers' acceptance of their appearance. Customers will subjectively think they

are interacting with socially animate beings, so the indifference of tangibility between the service principals therefore covers the mediation effect of friendliness.

6.2. Theoretical contributions

This study makes several theoretical contributions to the literature. First, it is one of the first to provide empirical evidence for the specific difference in SERVQUAL between a service robot and human staff. From the perspective of the service principal, we have enhanced the comparison of service quality between a service robot and human staff that was introduced in previous studies (e.g., Choi, Choi, Oh, & Kim, 2020). In contrast with previous studies that focus on general service quality, this study applies SERVQUAL to compare service quality, and the specific framework of SERVQUAL provides a clearer perspective for understanding the difference between robots and human employees.

Second, this study is the first to elucidate the key attributes of the service principal and to explain the mediation mechanism of the difference in service quality between a service robot and human staff through their attributes. While previous studies mention only scattered attributes of service principals, we identify five key attributes with the assistance of the seller-customer communication model developed by Sheth (1976). By applying these identified attributes, we are able to identify the driving mechanism of service quality difference between a service robot and human staff. Our study also helps to expand the limited body of research (e.g., Wirtz et al. (2018); Xiao and Kumar

(2021)) on service quality difference between a service robot and human staff, which has been lacking in empirical evidence.

Finally, our study extends MEC theory into robot service scenarios. It is therefore a meaningful response to the call to apply MEC theory in different service contexts (Blut et al., 2015; Orsingher et al., 2011). The proposed model and associated tests reveal that MEC theory is a useful tool for elucidating customers' evaluation of service quality through key attributes of different service principals. Thus, this study extends the scope of MEC theory and illustrates some new ways to apply MEC theory in service contexts.

6.3. Practical implications

We offer several practical implications. First, managers should pay attention to and make use of the strengths and weaknesses of service robots when introducing and managing robot services. Considering that service robots behave better in terms of stability and reliability, managers should enlist them to perform monotonous, dangerous, and repetitive tasks and extend their working hours. For example, hotel robots are available to carry luggage around the clock with consistent working status. Managers should also assign robots to perform tasks that tolerate only a low probability of service errors or tasks where it is necessary to save labor costs. Compared to human staff in terms of responsiveness and empathy, a service robot appears clumsy and even inept in communicating with customers (Hu et al., 2021). Therefore, in order to create positive interaction experiences for customers, managers should try to avoid designating robots to participate in service projects where they might need to respond to the personalized demands of customers. In addition, as there is no significant difference in tangibility between a service robot and human staff, the focus on a robot's appearance is currently excessive and may add unnecessary costs. These findings indicate that there is no need to highlight tangibility in applying a service robot, and they reveal the failure of many restaurants that blindly pursue a high degree of appearance anthropomorphism.

Second, enterprises and managers should take appropriate measures and make policies to effectively match the two service principals based on their attributes and advantages. Service enterprises can arrange human staff to assist service robots in some tasks that require high responsiveness and empathy. For example, human staff rather than robots should accompany customers to start services during their first visit to a robot hotel. In particular, human staff should help customers get familiar with the instructions associated with robot services. Human staff should always pay attention to the working status of service robots and correct their mistakes in time (such as moving the robots that deviate from the predetermined orbit to the correct path). Meanwhile, service robots can help human staff perform repetitive, tiring tasks that require high stability and reliability. For example, in the process of serving food, a service robot may carry dishes from the kitchen to different tables, and human staff only need to move the plate from the robot's hand onto the table. Due to the high reliability of a service robot's information system and data analysis, when a customer asks human staff for detailed service information (e.g., hotel room

information and scenic route information), a service robot can offer help and give a quick reply. Thus, through reasonable division of labor, the attributes of the two service principals can complement each other and service benefits can be maximized. Furthermore, enterprises should continually explore how a service robot and human staff perform joint complementary functions and assess the performance (or service quality) of the service principals using new standards (Fuentes-Moraleda et al., 2020). Along with these new standards, enterprises should develop different evaluation systems for service robots and human staff. For example, they should adopt specific rules for human staff who work with robots and create criteria to turn off or reduce the working hours of robots that behave badly.

Third, human staff need to face up to the state of robot service and improve their abilities to compete and cooperate with robots. To do this, they can increase and take advantage of their superior attributes (e.g., adaptability and creativity). For instance, human staff can enhance customers' interaction experience by meeting customers' various individualized, fuzzy and temporary needs. They can initiate services and discover customers' latent needs through empathy, creativity, and understanding of the service experience from the customer perspective. Human staff should also avoid or decrease their involvement in tasks that need a high degree of consistency and stability. To maintain sufficient stability, human staff should restrain their personal emotions and avoid being affected by the external environment. They should also recognize that it is necessary to work well with robots and adapt accordingly (Vatan & Dogan, 2021). If introducing a service robot is necessary and unavoidable, enterprises can train human staff to provide them with additional skills to enhance their motivation and self-efficacy when using a service robot.

6.4. Limitations and future directions

This study has several limitations. First, it only studies two types of service principal, which is slightly insufficient. We encourage researchers in this area to add more types of service principals in future studies, especially the combination of human staff and service robots. Second, this study only investigates the restaurant and hotel contexts. In order to gain an in-depth understanding of service quality in different scenarios, future research could explore different service settings (e.g., homes and medical facilities) where customers may have different service perceptions. Third, in addition to service principal and attribute, there may be other customer-related variables affecting service quality (e.g., customer's relationship orientation, technology readiness and robot service experience). Future studies should elucidate the influence of these variables.

Acknowledgements

The work was substantially supported by National Natural Science Foundation of China (No. 71701002) and Anhui Province Philosophy, Social Science Planning Project (No. AHSKY2017D24).

Appendix

1. Stimuli of study

In the experiment, the following two service scenarios were simulated:

Scenario 1 for human staff: One day, some of your friends come from far away to see you. At that time, you make a reservation by telephone for a restaurant (hotel) in your area. When you and your friends arrive at the restaurant (hotel), the staff person greets you and says "Welcome!" and guides you to the designated table (helps you check-in. After that, the human staff person explains hotel facilities and then leads you to your

room). In the dining area, the waiter provides you with the menu and guides you to order. Not long after you have successfully finished ordering, the waiter delivers your orders to the table.

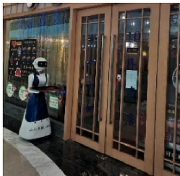
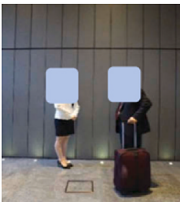
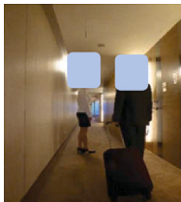
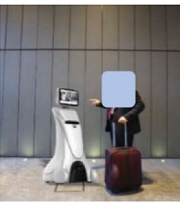
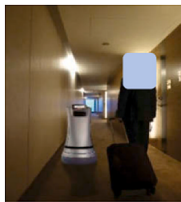
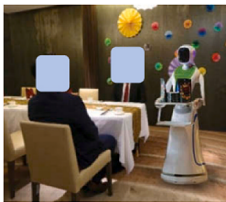
In scenario 1, researchers showed the images and service process of human staff in three service encounter contexts (Study 1 includes reception, taking orders and serving food; Study 2 includes concierge service at lobby,

hallway guidance and dining service). The images of human staff reflected real working employees, wearing uniforms and smiling and interacting with customers using their body movements.

Scenario 2 for the service robot: One day, some of your friends come from far away to see you. At that time, you make a reservation by telephone for a restaurant (hotel) in your area. When you and your friends arrive at the restaurant (hotel), the service robot greets you and says “Welcome!” After you arrive at the designated table (helps you check-in. After that, the service robot explains hotel facilities and then leads you to your room), you can order from the tablet interface carried by the service robot. Not long after your order is complete, the service robot delivers your orders to the table, and then you get up to accept your food.

In scenario 2, researchers showed the images and service process of the service robot in three service encounter contexts (Study 1 includes reception, taking orders and serving food; Study 2 includes concierge service, hallway guidance and dining service). The robot does not have a face or a smile, but it can communicate mechanically with customers, and it completes the service process according to customer requirements.

Stimuli of study.

	Service Setting		
	Reception	Ordering	Serving
Study 1 (Restaurant) Scenario 1 (Human staff; N = 151)			
Scenario 2 (Service robot; N = 154)			
	Lobby	Hallways	Dining
Study 2 (Hotel) Scenario 1 (Human staff; N = 112)			
Scenario 2 (Service robot; N = 110)			

II. Questionnaire items.

Variable	Observed variables	Items
Reliability (RE)	RE1	Staff can fulfil the service promise
	RE2	Staff can provide the service according to the service standard
	RE3	Staff can provide the service within the specified time
	RE4	No error occurs when the staff provides the service
Responsiveness (RES)	RES1	Staff is able to respond to customer requests in a timely manner
	RES2	Staff can provide services quickly
	RES3	Staff has the intention to serve
Assurance (AS)	AS1	Staff is trustworthy
	AS2	Staff has the ability to provide services

(continued on next page)

(continued)

Variable	Observed variables	Items
Empathy (EM)	AS3	The service provided by the staff is reassuring
	EM1	Staff can adopt the customer's point of view to consider the problem
	EM2	Staff can understand the individual needs of the customer
	EM3	Staff can try their best to meet the individual needs of the customer
Tangibility (TA)	EM4	Staff can discover the customer's personalized demands in time
	TA1	The image of the staff is acceptable
	TA2	The external image of the staff is appropriate
	TA3	The appearance of the staff is attractive
Stability (ST)	ST1	Staff does not make mistakes during service
	ST2	The service state and level of the staff are stable
	ST3	Staff can provide services without discrimination
	AD1	Staff can adapt to the customer's needs
Adaptability (AD)	AD2	Staff can adjust the service method according to the customer's situation
	AD3	Staff can provide the service in the way that the customer likes
	AB1	Staff pays much attention when providing services
	AB2	Staff can provide services without external interference
Absorption (AB)	AB3	Staff can immerse themselves in the state of service
	CR1	Staff can provide service creatively
	CR2	Staff is creative during the process of work
	CR3	Staff has innovative ideas and actions
Creativity (CR)	FR1	Staff can provide the service in a friendly manner
	FR2	Staff can kindly provide the service
	FR3	Staff has affinity

References

- Ab Rahman, N., Mohamed, M., & Sulong, F. (2022). Does covid-19 drive robot acceptance? An exploratory study of service robot in hospitality. *Tourism and Hospitality Management-Croatia*, 28(1), 193–209. <https://doi.org/10.20867/thm.28.1.10>
- Ackerman, E. (2016). *Study: Nobody wants social robots that look like humans because they threaten our identity*. <https://spectrum.ieee.org/automaton/robotics/humanoids/study-nobody-wants-social-robots-that-look-like-humans>.
- Ariffin, A. A. M., & Maghzi, A. (2012). A preliminary study on customer expectations of hotel hospitality: Influences of personal and hotel factors. *International Journal of Hospitality Management*, 31(1), 191–198. <https://doi.org/10.1016/j.ijhm.2011.04.012>
- Baron, R. M., & Kenny, D. A. (1986). The moderator-mediator variable distinction in social psychological research: Conceptual, strategic, and statistical considerations. *Journal of Personality and Social Psychology*, 51(6), 1173–1182. <https://doi.org/10.1037/0022-3514.51.6.1173>
- Bartneck, C., Kulic, D., Croft, E., & Zoghbi, S. (2009). Measurement instruments for the anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety of robots. *International Journal of Social Robotics*, 1(1), 71–81. <https://doi.org/10.1007/s12369-008-0001-3>
- Belanche, D., Casalo, L. V., Flavian, C., & Schepers, J. (2020). Service robot implementation: A theoretical framework and research agenda. *Service Industries Journal*, 40(3–4), 203–225. <https://doi.org/10.1080/02642069.2019.1672666>
- Berezina, K., Ciftci, O., & Cobanoglu, C. (2019). Robots, artificial intelligence, and service automation in restaurants. In S. Ivanov, & C. Webster (Eds.), *Robots, artificial intelligence and service automation in travel, tourism and hospitality* (pp. 185–219). Bingley, England: Emerald Publishing Limited. <https://doi.org/10.1108/JTF-06-2020-149>
- Blut, M., Chowdhry, N., Mittal, V., & Brock, C. (2015). E-Service quality: A meta-analytic review. *Journal of Retailing*, 91(4), 679–700. <https://doi.org/10.1016/j.jretai.2015.05.004>
- de Brentani, U. (2001). Innovative versus incremental new business services: Different keys for achieving success. *Journal of Product Innovation Management*, 18(3), 169–187. [https://doi.org/10.1016/S0737-6782\(01\)00071-6](https://doi.org/10.1016/S0737-6782(01)00071-6)
- Briscoe, J. P., Hall, D. T., & DeMuth, R. L. F. (2006). Protean and boundaryless careers: An empirical exploration. *Journal of Vocational Behavior*, 69(1), 30–47. <https://doi.org/10.1016/j.jvb.2005.09.003>
- Burns, E. C., Martin, A. J., & Collie, R. J. (2018). Adaptability, personal best (PB) goals setting, and gains in students' academic outcomes: A longitudinal examination from a social cognitive perspective. *Contemporary Educational Psychology*, 53, 57–72. <https://doi.org/10.1016/j.cedpsych.2018.02.001>
- Byrd, K., Fan, A., Her, E., Liu, Y., Almanza, B., & Leitch, S. (2021). Robot vs human: Expectations, performances and gaps in off-premise restaurant service modes. *International Journal of Contemporary Hospitality Management*. <https://doi.org/10.1108/ijchm-07-2020-0721>
- Caic, M., Mahr, D., & Oederkerken-Schroeder, G. (2019). Value of social robots in services: Social cognition perspective. *Journal of Services Marketing*, 33(4), 463–478. <https://doi.org/10.1108/jsm-02-2018-0080>
- Carpenter, J., Davis, J. M., Erwin-Stewart, N., Lee, T. R., Bransford, J. D., & Vye, N. (2009). Gender representation and humanoid robots designed for domestic use. *International Journal of Social Robotics*, 1(3), 261–265. <https://doi.org/10.1007/s12369-009-0016-4>
- Chand, M. (2010). Measuring the service quality of Indian tourism destinations: An application of SERVQUAL model. *International Journal of Services Technology and Management*, 13(3–4), 218–233. <https://doi.org/10.1504/ijstm.2010.032079>
- Chan, A. P. H., & Tung, V. W. S. (2019). Examining the effects of robotic service on brand experience: The moderating role of hotel segment. *Journal of Travel & Tourism Marketing*, 36(4), 458–468. <https://doi.org/10.1080/10548408.2019.1568953>
- Chen, Q., Gong, Y., Lu, Y., & Tang, J. (2022). Classifying and measuring the service quality of AI chatbot in frontline service. *Journal of Business Research*, 145, 552–568. <https://doi.org/10.1016/j.jbusres.2022.02.088>
- Chiang, A. H., & Trimi, S. (2020). Impacts of service robots on service quality. *Service Business*, 14(3), 439–459. <https://doi.org/10.1007/s11628-020-00423-8>
- Chi, O. H., Jia, S., Li, Y., & Gursoy, D. (2021). Developing a formative scale to measure consumers' trust toward interaction with artificially intelligent (AI) social robots in service delivery. *Computers in Human Behavior*, 118, Article 106700. <https://doi.org/10.1016/j.chb.2021.106700>
- Choi, S., Choi, C., & Mattila, A. S. (2020). Are all smiles perceived equal? The role of service provider's gender. *Service Science*, 12(1), 1–7. <https://doi.org/10.1287/serv.2019.0252>
- Choi, Y., Choi, M., Oh, M., & Kim, S. (2020). Service robots in hotels: Understanding the service quality perceptions of human-robot interaction. *Journal of Hospitality Marketing & Management*, 29(6), 613–635. <https://doi.org/10.1080/19368623.2020.1703871>
- Dane, E., & Brummel, B. J. (2013). Examining workplace mindfulness and its relations to job performance and turnover intention. *Human Relations*, 67(1), 105–128. <https://doi.org/10.1177/0018726713487753>
- Ding, K., Choo, W. C., Ng, K. Y., & Ng, S. I. (2020). Employing structural topic modelling to explore perceived service quality attributes in Airbnb accommodation. *International Journal of Hospitality Management*, 91. <https://doi.org/10.1016/j.ijhm.2020.102676>
- van Doorn, J., Mende, M., Noble, S. M., Hulland, J., Ostrom, A. L., Grewal, D., & Petersen, J. A. (2017). Domo Arigato Mr. Roboto: Emergence of automated social presence in organizational frontlines and customers' service experiences. *Journal of Service Research*, 20(1), 43–58. <https://doi.org/10.1177/1094670516679272>
- Elliott, K., Meng, G., & Hall, M. (2012). The influence of technology readiness on the evaluation of self-service technology attributes and resulting attitude toward technology usage. *Services Marketing Quarterly*, 33(4), 311–329. <https://doi.org/10.1080/15332969.2012.715049>
- Frese, M., Fay, D., Hilburger, T., Leng, K., & Tag, A. (1997). The concept of personal initiative: Operationalization, reliability and validity in two German samples. *Journal of Occupational and Organizational Psychology*, 70, 139–161.
- Fuentes-Moraleda, L., Díaz-Pérez, P., Orea-Giner, A., Mazón, A. M., & Villacé-Moliner, T. (2020). Interaction between hotel service robots and humans: A hotel-specific service robot acceptance model (sRAM). *Tourism Management Perspectives*. <https://doi.org/10.1016/j.tmp.2020.100751>
- Fugate, M., Kinicki, A. J., & Ashforth, B. E. (2004). Employability: A psycho-social construct, its dimensions, and applications. *Journal of Vocational Behavior*, 65(1), 14–38. <https://doi.org/10.1016/j.jvb.2003.10.005>
- Gardial, S. F., Clemons, D. S., Woodruff, R. B., Schumann, D. W., & Burns, M. J. (1994). Comparing consumers' recall of prepurchase and postpurchase product evaluation experiences. *Journal of Consumer Research*, 20(4), 548–560. <https://doi.org/10.1086/209369>

- Ghazizadeh, M., Lee, J. D., & Boyle, L. N. (2012). Extending the technology acceptance model to assess automation. *Cognition, Technology & Work*, 14(1), 39–49. <https://doi.org/10.1007/s10111-011-0194-3>
- Guizzo, E. (2010). Hiroshi Ishiguro: The man who made a copy of himself. <https://spectrum.ieee.org/hiroshi-ishiguro-the-man-who-made-a-copy-of-himself>.
- Gutman, J. (1982). A means-end chain model based on consumer categorization processes. *Journal of Marketing*, 46(2), 60–72. <https://doi.org/10.2307/3203341>
- Hayes, A. F., & Preacher, K. J. (2014). Statistical mediation analysis with a multicategorical independent variable. *British Journal of Mathematical and Statistical Psychology*, 67(3), 451–470. <https://doi.org/10.1111/bmsp.12028>
- Heider, & Fritz. (1958). *The psychology of interpersonal relationships*. Psychology Press.
- Hengstler, M., Enkel, E., & Duelli, S. (2016). Applied artificial intelligence and trust-The case of autonomous vehicles and medical assistance devices. *Technological Forecasting and Social Change*, 105, 105–120. <https://doi.org/10.1016/j.techfore.2015.12.014>
- Hess, R. L., Ganesan, S., & Klein, N. M. (2003). Service failure and recovery: The impact of relationship factors on customer satisfaction. *Journal of the Academy of Marketing Science*, 31(2), 127–145. <https://doi.org/10.1177/0092070302250898>
- Ho, T. H., Tojib, D., & Tsarenko, Y. (2020). Human staff vs. service robot vs. fellow customer: Does it matter who helps your customer following a service failure incident? *International Journal of Hospitality Management*, 87. <https://doi.org/10.1016/j.ijhm.2020.102501>. Article 102501.
- Hou, Y., Zhang, K., & Li, G. (2021). Service robots or human staff: How social crowding shapes tourist preferences *Tourism Management* (Vol. 83). <https://doi.org/10.1016/j.tourman.2020.104242>
- Huang, M. H., & Rust, R. T. (2021). Engaged to a robot? The role of AI in service. *Journal of Service Research*, 24(1), 30–41. <https://doi.org/10.1177/1094670520902266>, 1094670520902266.
- Hu, Y., Min, H., & Su, N. (2021). How sincere is an apology? Recovery satisfaction in a robot service failure context. *Journal of Hospitality & Tourism Research*, 45(6), 1022–1043. <https://doi.org/10.1177/10963480211011533>. Article 10963480211011533.
- Ivanov, S., Seyitoglu, F., & Markova, M. (2020). Hotel managers' perceptions towards the use of robots: A mixed-methods approach. *Information Technology & Tourism*, 22(4), 505–535. <https://doi.org/10.1007/s40558-020-00187-x>
- Ivanov, S., Webster, C., & Seyyedi, P. (2018). Consumers' attitudes towards the introduction of robots in accommodation establishments. *Tourism*, 66(3), 302–317.
- van Iwaarden, J., van der Wiele, T., Ball, L., & Millen, R. (2003). Applying SERVQUAL to Web sites: An exploratory study. *International Journal of Quality & Reliability Management*, 20(8), 919–935. <https://doi.org/10.1108/02656710310493634>
- Jiang, J. J., Klein, G., & Carr, C. L. (2002). Measuring information system service quality: Servqual from the other side. *MIS Quarterly*, 26(2), 145–166. <https://doi.org/10.2307/4132324>
- Jiang, Y., & Wen, J. (2020). Effects of COVID-19 on hotel marketing and management: A perspective article. *International Journal of Contemporary Hospitality Management*, 32(8), 2563–2573. <https://doi.org/10.1108/ijchm-03-2020-0237>
- Jones, K. S., & Schmidlin, E. A. (2011). Human-robot interaction: Toward useable personal service robots. *Reviews of Human Factors & Ergonomics*, 7(1), 100–148.
- Kabadayi, S., Ali, F., Choi, H., Joosten, H., & Lu, C. (2019). Smart service experience in hospitality and tourism services: A conceptualization and future research agenda. *Journal of Service Management*, 30(3), 326–348. <https://doi.org/10.1108/josm-11-2018-0377>
- Kano, N., Seraku, N., Takahashi, F., & Tsuji, S. (1984). Attractive quality and must-be quality. *Journal of Japanese Society for Quality Control*, 14, 147–156.
- Keeling, K., McGoldrick, P., & Beatty, S. (2010). Avatars as salespeople: Communication style, trust, and intentions. *Journal of Business Research*, 63(8), 793–800. <https://doi.org/10.1016/j.jbusres.2008.12.015>
- Kim, S., Kim, J., Badu-Baiden, F., Giroux, M., & Choi, Y. (2021). Preference for robot service or human service in hotels? Impacts of the COVID-19 pandemic. *International Journal of Hospitality Management*, 93. <https://doi.org/10.1016/j.ijhm.2020.102795>
- Kim, Y., & Lee, H. S. (2014). Quality, perceived usefulness, user satisfaction, and intention to use: An empirical study of ubiquitous personal robot service. *Asian Social Science*, 10(11), 1–16. <https://doi.org/10.5539/ass.v10n11p1>
- Köhler, C. F., Rohm, A. J., Ruyter, K.d., & Wetzels, M. (2011). Return on interactivity: The impact of online agents on newcomer adjustment. *Journal of Marketing*, 75(2), 93–108. <https://doi.org/10.1509/jm.75.2.93>
- Kotler, P. (1994). *Marketing management : Analysis, planning, implementation, and control* (8th ed.). Prentice-Hall International, Inc.
- Kumar, V., Rajan, B., Gupta, S., & Pozza, I. D. (2019). Customer engagement in service. *Journal of the Academy of Marketing Science*, 47(1), 138–160. <https://doi.org/10.1007/s11747-017-0565-2>
- Lai, T. L. (2004). Service quality and perceived value's impact on satisfaction, intention and usage of short message service (SMS). *Information Systems Frontiers*, 6(4), 353–368. <https://doi.org/10.1023/B:ISIF.0000046377.32617.3d>
- Lee, G., & Kim, Y. (2022). Effects of resistance barriers to service robots on alternative attractiveness and intention to use. *Sage Open*, 12(2), Article 21582440221099293. <https://doi.org/10.1177/21582440221099293>
- Lee, W. J., Kwag, S. I., & Ko, Y. D. (2020). Optimal capacity and operation design of a robot logistics system for the hotel industry. *Tourism Management*, 76, Article 103971. <https://doi.org/10.1016/j.tourman.2019.103971>
- Lin, H., Chi, O. H., & Gursoy, D. (2020). Antecedents of customers' acceptance of artificially intelligent robotic device use in hospitality services. *Journal of Hospitality Marketing & Management*, 29(5), 530–549. <https://doi.org/10.1080/19368623.2020.1685053>
- Lin, I. Y., & Mattila, A. S. (2021). The value of service robots from the hotel guest's perspective: A mixed-method approach. *International Journal of Hospitality Management*, 94, Article 102876. <https://doi.org/10.1016/j.ijhm.2021.102876>
- Lin, S. P., Yang, C. L., Chan, Y. H., & Sheu, C. (2010). Refining Kano's 'quality attributes-satisfaction' model: A moderated regression approach. *International Journal of Production Economics*, 126(2), 255–263. <https://doi.org/10.1016/j.ijpe.2010.03.015>
- Li, Y. N., Tan, K. C., & Xie, M. (2002). Measuring web-based service quality. *Total Quality Management*, 13(5), 685–700. <https://doi.org/10.1080/0954412022000002072>
- Lu, L., Zhang, P., & Zhang, T. (2021). Leveraging "human-likeness" of robotic service at restaurants. *International Journal of Hospitality Management*, 94, Article 102823. <https://doi.org/10.1016/j.ijhm.2020.102823>
- Ming, L., Li, Y., Peng, Q., Wang, J., & Yu, C. (2021). Evaluating community question-answering websites using interval-valued intuitionistic fuzzy DANP and TODIM methods. *Applied Soft Computing*, 99. <https://doi.org/10.1016/j.asoc.2020.106918>
- Molly, M. (2016). Useless robot waiters fired for incompetence in China. <http://www.telegraph.co.uk/technology/2016/04/11/useless-robot-waiters-fired-for-incompetence-in-china/>.
- Mori, M. (2012). The uncanny valley. *IEEE Robotics and Automation Magazine*, 19(2), 98–100. <https://doi.org/10.1109/mra.2012.2192811>
- Morita, T., Kashiwagi, N., Yorozu, A., Suzuki, H., & Yamaguchi, T. (2018, Sep 03–05). Implementing multi-robot cafe by PRINTEPS with service quality dimensions. In *Procedia Computer Science 22nd international Conference on knowledge-based and intelligent information and Engineering systems (KES)*, Belgrade, Serbia.
- Murphy, J., Gretzel, U., & Pesonen, J. (2019). Marketing robot services in hospitality and tourism: The role of anthropomorphism. *Journal of Travel & Tourism Marketing*, 36(7), 784–795. <https://doi.org/10.1080/10548408.2019.1571983>
- Murphy, J., Hofacker, C., & Gretzel, U. (2017). Dawning of the age of robots in hospitality and tourism: Challenges for teaching and research. *European Journal of Tourism Research*, 15, 104–111.
- Nowak, A., Vallacher, R. R., & Zochowski, M. (2005). The emergence of personality: Dynamic foundations of individual variation. *Developmental Review*, 25(3–4), 351–385. <https://doi.org/10.1016/j.dr.2005.10.004>
- Oghuma, A. P., Fernando Libaque-Saenz, C., Wong, S. F., & Chang, Y. (2016). An expectation-confirmation model of continuance intention to use mobile instant messaging. *Telematics and Informatics*, 33(1), 34–47. <https://doi.org/10.1016/j.tele.2015.05.006>
- Orsingher, C., Marzocchi, G. L., & Valentini, S. (2011). Consumer (goal) satisfaction: A means-ends chain approach. *Psychology and Marketing*, 28(7), 730–748. <https://doi.org/10.1002/mar.20409>
- Paiva, A., Leite, I., Boukricha, H., & Wachsmuth, I. (2017). Empathy in virtual agents and robots: A survey. *Acm Transactions on Interactive Intelligent Systems*, 7(3), 11. <https://doi.org/10.1145/2912150>
- Parasuraman, A., Zeithaml, V., & Berry, L. (1988). Servqual: A multiple-item scale for measuring consumer perceptions of service quality. *Journal of Retailing*, 61(1), 12–40.
- Pozharliev, R., De Angelis, M., Rossi, D., Romani, S., Verbeke, W., & Cherubino, P. (2021). Attachment styles moderate customer responses to frontline service robots: Evidence from affective, attitudinal, and behavioral measures. *Psychology and Marketing*, 38(5), 881–895. <https://doi.org/10.1002/mar.21475>
- Prentice, C., & Nguyen, M. (2021). Robotic service quality – scale development and validation. *Journal of Retailing and Consumer Services*, 62. <https://doi.org/10.1016/j.jretconser.2021.102661>
- Price, L. L., Arnold, E. J., & Deibler, S. L. (1995). Consumers' emotional responses to service encounters. *International Journal of Service Industry Management*, 6(3), 34–63. <https://doi.org/10.1108/09564239510091330>
- Qiu, H., Li, M., Shu, B., & Bai, B. (2020). Enhancing hospitality experience with service robots: The mediating role of rapport building. *Journal of Hospitality Marketing & Management*, 29(3), 247–268. <https://doi.org/10.1080/19368623.2019.1645073>
- Reese, B. (2018). *The fourth age: Smart robots, conscious computers, and the future of humanity*. Simon and Schuster.
- Richard, S., & Mansfield. (1996). Building competency models: Approaches for HR professionals. *Human Resource Management*, 35(1), 7–18. [https://doi.org/10.1002/\(sici\)1099-050x\(199621\)35:1<7::aid-hrm1>3.0.co;2-2](https://doi.org/10.1002/(sici)1099-050x(199621)35:1<7::aid-hrm1>3.0.co;2-2)
- Richardson, H. A., Simmering, M. J., & Sturman, M. C. (2009). A tale of three perspectives. *Organizational Research Methods*, 12(4), 762–800. <https://doi.org/10.1177/1094428109332834>
- Rosete, A., Soares, B., Salvadorinho, J., Reis, J., & Amorim, M. (2020). Service robots in the hospitality industry: An exploratory literature review. In *Exploring service Science* (pp. 174–186). https://doi.org/10.1007/978-3-030-38724-2_13
- Schaufeli, W., & Bakker, A. (2001). Work and well-being: Towards a positive occupational health psychology. *Gedrag & Organization*, 14(5), 229–253.
- Schaufeli, W., Salanova, M., González-romá, V., & Bakker, A. (2002). The measurement of engagement and burnout: A two sample confirmatory factor analytic approach. *Journal of Happiness Studies*, 3(1), 71–92. <https://doi.org/10.1023/a:1015630930326>
- Schepers, J., & Streukens, S. (2022). To serve and protect: A typology of service robots and their role in physically safe services. *Journal of Service Management*. <https://doi.org/10.1108/josm-11-2021-0409>
- Shah, F. T., Syed, Z., Imam, A., & Raza, A. (2020). The impact of airline service quality on passengers' behavioral intentions using passenger satisfaction as a mediator. *Journal of Air Transport Management*, 85. <https://doi.org/10.1016/j.jairtraman.2020.101815>
- Sheth, J. N. (1976). Buyer-seller interaction: A conceptual framework. *Advances in Consumer Research*, 3(1).

- Shin, H. H., & Jeong, M. (2020). Guests' perceptions of robot concierge and their adoption intentions. *International Journal of Contemporary Hospitality Management*, 32(8), 2613–2633. <https://doi.org/10.1108/ijchm-09-2019-0798>
- Singh, A., & Prasher, A. (2017). Measuring healthcare service quality from patients' perspective: Using Fuzzy AHP application. *Total Quality Management and Business Excellence*, 30(3–4), 284–300. <https://doi.org/10.1080/14783363.2017.1302794>
- Sonnentag, S., & Krueger, U. (2006). Psychological detachment from work during off-job time: The role of job stressors, job involvement, and recovery-related self-efficacy. *European Journal of Work & Organizational Psychology*, 15(2), 197–217. <https://doi.org/10.1080/13594320500513939>
- Stanton, W. J., Etzel, & Michael, J. (1991). *Fundamentals marketing* (9th ed.).
- Tan, K. C., & Pawitra, T. A. (2001). Integrating SERVQUAL and Kano's model into QFD for service excellence development. *Managing Service Quality*, 11(6), 418–430. <https://doi.org/10.1108/EUM00000000006520>
- Teng, C. L., & Liu, T. W. (2014). How do personality interactions affect service quality? The perspective of processing efficiency theory. *Service Business*, 8(2), 375–397. <https://doi.org/10.1007/s11628-013-0200-3>
- Tierney, P., Farmer, S. M., & Graen, G. B. (1999). An examination of leadership and employee creativity: The relevance of traits and relationships. *Personnel Psychology*, 52(3), 591–620. <https://doi.org/10.1111/j.1744-6570.1999.tb00173.x>
- Tontini, G., Corrêa, H., & Silveira, A. (2007). Identification of satisfaction attributes using competitive analysis of the improvement gap. *International Journal of Operations & Production Management*, 27(5), 482–500. <https://doi.org/10.1108/01443570710742375>
- Tsai, W. C., & Huang, Y. M. (2002). Mechanisms linking employee affective delivery and customer behavioral intentions. *Journal of Applied Psychology*, 87(5), 1001–1008. <https://doi.org/10.1037/0021-9010.87.5.1001>
- Tuli, K. R., Kohli, A. K., & Bharadwaj, S. G. (2007). Rethinking customer solutions: From product bundles to relational processes. *Journal of Marketing*, 71(3), 1–17. <https://doi.org/10.1509/jmkg.71.3.1>
- Tung, V. W. S., & Au, N. (2018). Exploring customer experiences with robotics in hospitality. *International Journal of Contemporary Hospitality Management*, 30(7), 2680–2697. <https://doi.org/10.1108/ijchm-06-2017-0322>
- Tung, V. W. S., & Law, R. (2017). The potential for tourism and hospitality experience research in human-robot interactions. *International Journal of Contemporary Hospitality Management*, 29(10), 2498–2513. <https://doi.org/10.1108/ijchm-09-2016-0520>
- Tuomi, A., Tussyadiah, I. P., & Stienmetz, J. (2021). Applications and implications of service robots in hospitality. *Cornell Hospitality Quarterly*, 62(2), 232–247. <https://doi.org/10.1177/1938965520923961>. Article 1938965520923961.
- Tzeng, G. H., & Chang, H. F. (2011). Applying importance-performance analysis as a service quality measure in food service industry. *Journal of Technology Management and Innovation*, 6(3), 106–115. <https://doi.org/10.4067/s0718-27242011000300008>
- Tzou, J. H., & Su, K. L. (2009). High-speed laser localization for a restaurant service mobile robot. *Artificial Life and Robotics*, 14(2), 252–256. <https://doi.org/10.1007/s10015-009-0665-2>
- Vatan, A., & Dogan, S. (2021). What do hotel employees think about service robots? A qualitative study in Turkey. *Tourism Management Perspectives*, 37. <https://doi.org/10.1016/j.tmp.2020.100775>
- Verhagen, T., van Nes, J., Feldberg, F., & van Dolen, W. (2014). Virtual customer service agents: Using social presence and personalization to shape online service encounters. *Journal of Computer-Mediated Communication*, 19(3), 529–545. <https://doi.org/10.1111/jcc4.12066>
- Wieseke, J., Geigenmueller, A., & Kraus, F. (2012). On the role of empathy in customer-employee interactions. *Journal of Service Research*, 15(3), 316–331. <https://doi.org/10.1177/1094670512439743>
- Wirtz, J., Patterson, P. G., Kunz, W. H., Gruber, T., Lu, V. N., Paluch, S., & Martins, A. (2018). Brave new world: Service robots in the frontline. *Journal of Service Management*, 29(5), 907–931. <https://doi.org/10.1108/josm-04-2018-0119>
- Wuenderlich, N. V. (2009). *Acceptance of remote services*. Wiesbaden: Gabler.
- Wu, Y. H., Wrobel, J., Cornuet, M., Kerherve, H., Damnee, S., & Rigaud, A. S. (2014). Acceptance of an assistive robot in older adults: A mixed-method study of human-robot interaction over a 1-month period in the living lab setting. *Clinical Interventions in Aging*, 9, 801–811. <https://doi.org/10.2147/cia.S56435>
- Xiao, L., & Kumar, V. (2021). Robotics for customer service: A useful complement or an ultimate substitute? *Journal of Service Research*, 24(1), 9–29. <https://doi.org/10.1177/1094670519878881>. Article 1094670519878881.
- Xu, X. Y., Wu, S.-Y., & Jing, H. (2017). Explore the differences between perceived service quality and customer satisfaction based on customer expectation of service attributes. *International Journal of Services Technology and Management*, 23(3), 219–236. <https://doi.org/10.1504/IJSTM.2017.10006357>
- Yoganathan, V., Osburg, V.-S., Kunz, W. H., & Toporowski, W. (2021). Check-in at the Robo-desk: Effects of automated social presence on social cognition and service implications. *Tourism Management*, 85, Article 104309. <https://doi.org/10.1016/j.tourman.2021.104309>
- Yu, C. E., & Ngan, H. F. B. (2019). The power of head tilts: Gender and cultural differences of perceived human vs human-like robot smile in service. *Tourism Review*, 74(3), 428–442. <https://doi.org/10.1108/tr-07-2018-0097>
- Zalama, E., García-Bermejo, J. G., Marcos, S., Domínguez, S., Feliz, R., Pinillos, R., & López, J. (2014). Sacarino, a service robot in a hotel environment. In *ROBOT2013: First Iberian robotics Conference* (pp. 3–14). https://doi.org/10.1007/978-3-319-03653-3_1
- Zeithaml, V. A., Bitner, M. J., & McGrawHill. (2008). *Services marketing: Integrating customer focus across the firm*. China Machine Press.
- Zhang, X., Balaji, M. S., & Jiang, Y. (2022). Robots at your service: Value facilitation and value co-creation in restaurants. *International Journal of Contemporary Hospitality Management*. <https://doi.org/10.1108/ijchm-10-2021-1262>
- Zhou, J., & George, J. M. (2001). When job dissatisfaction leads to creativity: Encouraging the expression of voice. *Academy of Management Journal*, 44(4), 682–696. <https://doi.org/10.2307/3069410>
- Zhu, D. H., & Chang, Y. P. (2020). Robot with humanoid hands cooks food better? *International Journal of Contemporary Hospitality Management*, 32(3), 1367–1383. <https://doi.org/10.1108/IJCHM-10-2019-0904>